

Hints & Solutions

1. The energy of the projectile will be equivalent to potential energy of the charge system.

$$E = \frac{1}{4\pi\epsilon_0} \frac{(Z_1 e)(Z_2 e)}{r_0}$$

Energy $[E \propto Z_1 Z_2]$

2. According to Bohr's hypothesis, electron can revolve only in those orbits in which its angular

momentum is an integral multiple of $\frac{h}{2\pi}$; where

h is planck constant. This is called quantisation of angular momentum.

3. According to Bohr's theory

$$\text{We know } mvr = n \frac{h}{2\pi} \Rightarrow [mvr \propto n]$$

$$mv = n \frac{h}{2\pi r}$$

$$\text{Linear momentum } p = mv = \frac{nh}{2\pi r}$$

$$p \propto \frac{n}{r}$$

As $r \propto n^2$

$$\text{Hence } p \propto \frac{1}{n}$$

$$\text{We know } KE \propto \frac{1}{r} \text{ and } r \propto n^2$$

$$KE \propto \frac{1}{n^2}$$

4. In the second excited state $n = 3$

$$L_H = L_{Li} = 3 \left(\frac{h}{2\pi} \right)$$

$$Z_H = 1; Z_{Li} = 3; E \propto Z^2$$

$$|E_{Li}| = 9|E_H|$$

$$\text{or } |E_H| < |E_{Li}|$$

5. The minimum energy to ionise an atom is the energy required to excite an electron in the atom from its ground state to its ionisation level corresponding to $n = \infty$.

6. As electron makes a transition from an excited state to a ground state, n decrease there for kinetic energy increase but potential energy and total energy decrease.

$$\text{K.E.} = \frac{e^2}{8\pi\epsilon_0 n^2 a_0}$$

$$\text{P.E.} = \frac{-e^2}{4\pi\epsilon_0 n^2 a_0}$$

$$\text{T.E.} = \frac{-e^2}{8\pi\epsilon_0 n^2 a_0}$$

$$7. \quad mvR = \frac{nh}{2\pi} \quad \dots(1)$$

$$qvB = \frac{mv^2}{R}; \quad qB = \frac{mv}{R} \quad \dots(2)$$

from Eq. (1) & (2)

$$qB \left(\frac{nh}{2\pi mv} \right) = mv$$

$$\frac{1}{2} mv^2 = \frac{1}{4\pi m} nhqB$$

$$E = n \left[\frac{hqB}{4\pi m} \right]$$

$$8. \quad \frac{mv^2}{r} = \frac{3q^2}{4\pi\epsilon_0 r^2}$$

$$mvr = \frac{3q^2}{4\pi\epsilon_0 v} \quad \dots(1)$$

$$\text{and } \frac{nh}{2\pi} = mvr$$

using (1) & (2) and putting $[n = 1]$

$$\frac{h}{2\pi} = \frac{3q^2}{4\pi\epsilon_0 v}$$

$$v = \frac{3q^2}{2\epsilon_0 h}$$

9. As the collision is inelastic, it means a part of K.E. is transformed into some other form due to collision. In this case, the KE of incident electron can be absorbed by H atom and it can absorb only 10.2 eV out of 11.2 eV, so that it can reach to 1st excited state and electron leaves with remaining energy i.e. 1.0 eV.

10. $E = Rhc \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

(II) $E_{4-3} = Rhc \left[\frac{1}{3^2} - \frac{1}{4^2} \right] = Rhc \left[\frac{7}{9 \times 16} \right]$

$E_{[4-3]} = 0.05 Rhc$

(IV) $E_{[4-2]} = Rhc \left[\frac{1}{2^2} - \frac{1}{4^2} \right] = 0.2 Rhc$

(III) $E_{[2-1]} = Rhc \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = 0.75 Rhc$

(I) $E_{[1-4]} = Rhc \left[\frac{1}{4^2} - \frac{1}{1^2} \right] = -0.9 Rhc$

(it is absorption not emission)

11. For transition from $4E$ to E

$(4E - E) = \frac{hc}{\lambda_1} \Rightarrow \lambda_1 = \frac{hc}{3E} \quad \dots(1)$

For transition from $\frac{7}{3}E$ to E

$\frac{7}{3}E - E = \frac{hc}{\lambda_2} \Rightarrow \lambda_2 = \frac{3hc}{4E} \quad \dots(2)$

Thus $\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{4}{9}$

12. Masses of ${}_1H^1$ and ${}_1D^2$ are different, Hence the corresponding wavelengths are different.

$\frac{1}{\lambda} = Rz^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

3. Radius of n^{th} orbit $r_n \propto n^2$
graph between r_n and n is a parabola. Also

$\frac{r_n}{r_1} = \left(\frac{n}{1} \right)^2 \Rightarrow \log_e \left(\frac{r_n}{r_1} \right) = 2 \log_e(n)$

Comparing this equation with $y = mx + c$

graph between $\log_e \left(\frac{r_n}{r_1} \right)$ and $\log_e(n)$ will be a

straight line, passing from origin.

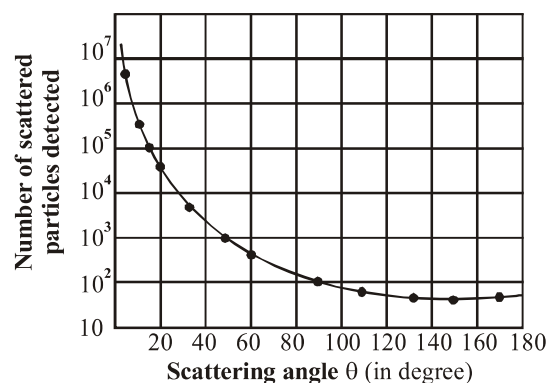
Time period (T_n) $\propto n^3$

14. Hydrogen atom is one electron revolving around a proton. A hydrogen ion is obtained when electron is removed.

15. Coulomb's electrostatic force of attraction between nucleus and electron supplies the necessary centripetal forces.

31. By the nineteenth century, enough evidence had accumulated in favour of atomic hypothesis of matter. In 1897, the experiments on electric discharge through gases carried out by the English physicist JJ Thomson (1856-1940) revealed that atoms of different elements contain negatively charged constituents (electrons) that are identical for all atoms.

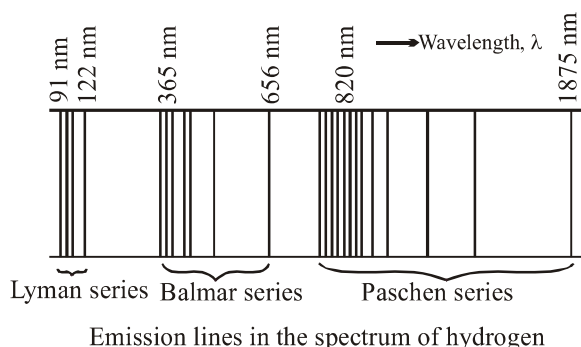
- 32.



The dots are α -particle scattering data for a thin gold foil obtained by Geiger and Marsden. The solid curve is the theoretical prediction based on the assumption that atom has a small dense and positively charged nucleus.

33. Correct option is '4' as most of α -particles pass straight through, few are deflected at small angles and very few are deflected by larger angles.
34. Rutherford atomic model is also called nuclear model as through this model nucleus as a point charge entity with whole mass was described first time.

35. A line emission spectrum is unique, just like a finger print. Line emission spectrum consists of bright lines on a dark background. The spectrum emitted by atomic hydrogen is shown in figure. The study of emission line spectra of a material can therefore serve as a type of 'fingerprint' for identification of the gas.



36. As the electrons spiral inwards, their angular velocities and hence their frequencies would change continuously, and so will the frequency of the light emitted. Thus, they would emit a continuous spectrum, in contradiction to the line spectrum actually observed. Clearly, Rutherford model tells only a part of the story implying that the classical ideas are not sufficient to explain the atomic structure.

37. As we know, orbital speed $v_n \propto \frac{1}{n}$ and frequency of revolution of an electron $f \propto \frac{1}{n^3}$.

Hence, it conclude that orbiting speed of an electron decreases as it shift to discrete as orbit away from nucleus and binding force of an electron towards nucleus increases as electron shift to other orbits.

38. In Bohr's model angular momentum of revolving electron is

$$L = n \frac{h}{2\pi}$$

So, it is quantised.

39. For an electron by de-Broglie's hypothesis $n\lambda = 2\pi r$

In ground state, $n = 1$

$\therefore \lambda = 2\pi r = \text{circumference of orbit.}$

40. As electrons are not subjected to a central force. So, Bohr's model cannot be applied to a multi-electron atom.

41. Here, $E_n = -\frac{13.6}{n^2} \text{ eV}$

$U_n = \text{Potential energy}$

$$2E_n = -\frac{27.2}{n^2} \text{ eV}$$

$$\text{and } K_n = |E_n| = \text{Kinetic energy} = \frac{13.6}{n^2} \text{ eV}$$

Clearly, or n increases PE and total energy being negative increases, where as KE decreases.

42. Bohr's second postulate defines these stable orbits. This postulate states that the electron revolves around the nucleus only in those orbits for which the angular momentum is some integral multiple of $h/2\pi$, where h is Planck's constant ($=6.6 \times 10^{-34} \text{ J-s}$). Thus, the angular momentum (L) of the orbiting electron is quantised i.e., $L = nh/2\pi$.

43. A set of atoms in an excited state decays in general to any of the state emits a photon and achieves lower energy state.

44. When white light passes through a gas and we analyse the transmitted light using a spectrometer, we find some dark lines in the spectrum. These dark lines correspond precisely to those wavelengths which were found in the emission line spectrum of the gas. This is called the absorption spectrum of the material of the gas.

45. In an absorption spectrum, dark lines corresponds to same wavelength which were found in emission line spectrum for all gases. Given statement is correct for all gases.